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Location allocation approach based on influence of points of dispensing

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Public health emergencies, such as disease outbreaks, require health authorities to set up points of dispensing (PODs) to efficiently distribute vaccines and drugs to attempt to reduce infection spread within a short timeframe. A considerable amount of research has been conducted to determine the number and location of new PODs while keeping existing ones operational. A query-based approach needs data on the number of new PODs in advance, and an optimisation-based approach requires the specification of distance and percentage gap as constraints or objective to ensure the required population coverage is achieved. The model presented here overcomes such issues. It takes an influence-based approach for POD location-allocation (IPLA) based on a new Influence Affinity Propagation (IAP) clustering algorithm to designate the number and location of new PODs where PODs already exist. IAP takes the influence of existing PODs as a novel weight, that is, the total population size attracted by existing PODs with minimum distance, in addition to geographical information. The proposed model is applied to three synthetic datasets and a real case, namely: the COVID-19 pandemic in Jeddah, Saudi Arabia. IPLA demonstrates better population coverage with a lower number of new PODs than other approaches. It also obtained the closest results to the case study in terms of number, location and predictive performance.

The setting up of testing and vaccination facilities is essential during public health emergencies, such as disease outbreaks, so that the population can be vaccinated against illness and necessary preventative measures can be put in place as quickly as possible to minimise the impact of the outbreak. These Points of Dispensing (PODs), as they are known, are the standard approach to mass dispensing of vaccines and drugs to demand points in a short timeframe[1,2]. PODs differ from regular healthcare facilities in that they provide medical supplies in an emergency, thus playing a critical role in protecting public health[3]. Determining the precise number and optimal location of PODs is vital in the drawing up of a response plan. Planning must take into account community contexts, such as the extent to which sites are available and the population density of a given area[4]. It is thus critical to select the best locations for PODs and to determine how the demand points can be efficiently allocated to PODs when creating a response plan, which is known as location-allocation problem.

The World Health Organization (WHO) declared the Coronavirus disease (COVID-19) to be a global pandemic in 2020[5]. During the COVID-19 pandemic, the Ministry of Health (MOH) in Saudi Arabia formulated a response plan for Jeddah city in 2020 that initially had 12 PODs; this number was increased to 42 PODs in 2021[6,7]. This could have been due to several different reasons, such as the need to cover more of the population and to provide easy access to PODs. Establishing new facility locations when there are existing facilities in place during emergencies is an important task[8]. Location selection queries (LSQs) and facility location problems (FLPs) are two approaches that are taken to identify the best locations for new facilities in relation to existing facilities.

FLPs aim to optimise certain values, such as minimising the number of new facilities to obtain predetermined coverage within a defined travel distance. The LSQs are filed in a spatial database and are used FLPs[9–13] to determine the potential location of a candidate facility that would place it closer to demand points than existing facilities. Such queries optimise specific functions by considering the potential location of a new facility within the context of existing facilities to minimise the average distance between demand points and the new facility. The existing LSQ-based solutions include sorted lists of candidate locations that are most attractive in terms of population or demand points based on minimum distance. Top t candidate locations are selected from the sorted list (e.g., t=3), which is known as top influence location selection (TILS).

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Current solutions, such as solving set covering location problems using the genetic algorithm (SCP-GA), based on FLPs to upgrade the number and locations of PODs, require advanced knowledge of the percentage coverage of the population and travelling times[8]. Additionally, TILS, which is based on a query-based approach, requires prior knowledge of the number of top PODs that are attracted by the population[14]. The proposed approach in[15] uses population affinity propagation (PAP) clustering to address the need for distance and population coverage in advance but it does not take into account the PODs that are already providing services. It is designed to initiate the emergency response plan and add POD locations at the outset of the plan. This is the main limitation of this approach: locations cannot be added later to the plan. Also, taking population as the sole independent factor to establish how the number of PODs that are needed is not sufficient as this approach does not take existing PODs into consideration. It is imperative that existing PODs be considered as a factor when locating new PODs. This research highlights the need for improvement in establishing both the number and locations of PODs that are needed in an emergency situation. It poses the following key research questions: What is the most appropriate location-allocation approach for upgrading the number and location of PODs to improve population coverage and provide easy access without predefined inputs such as the top number of PODs? Can the influence of existing PODs be used as a design factor to determine the number and location of new PODs?

To address this issue, an influence-based approach to POD location allocation (IPLA) is proposed for POD location allocation where there are existing PODs. Furthermore, a new weight is proposed, which is being referred to as 'influence of existing PODs'. The influence affinity propagation (IAP) clustering algorithm is also proposed based on the aforementioned new weight for designating the number and location of new PODs. IPLA consists of two phases. The first phase includes the designation of the number and location of new PODs. IPLA calculates the influence of existing PODs as the novel weight, that is, the total population that is attracted by existing PODs based on the minimum distance. Then, different pre-processing steps are employed, namely, normalisation and coordinates conversion. The proposed IAP algorithm uses the weight of existing PODs as a factor in identifying the new number and locations of PODs. Finally, locations are generated for the new PODs based on how far they are from existing POD locations with different distance thresholds. In the second phase, demand points are assigned and allocated to the closest existing and new PODs. The main contributions of this work are listed below:

- Influence-based approach to POD location allocation (IPLA) is proposed for POD location allocation where there are existing PODs.
- A novel weight is proposed, referred to as 'influence of existing PODs.'
- The influence affinity propagation (IAP) clustering algorithm is proposed based on the novel weight, that is, the influence of existing PODs on designating the number and location of new PODs.

Related works

This section presents existing work on locating new facilities/PODs where facilities/PODs already exist based on location selection queries and facility location problems.

Location selection query

The location selection query is filed in a spatial database. It is based on reverse nearest neighbour (RNN), which is a one-dimensional query that identifies a set of data points closest to the query point[16,17]. In contrast, the bichromatic reverse nearest neighbour (BRNN) query is a two-dimensional concept that has applications in many real-life domains[18,19]. It involves two types of data points, such as demand points and facilities, and aims to identify the nearest demand points to a facility.

In selecting the location of a given facility where other facilities are already in place these queries optimise specific functions. This includes min-distance and max-influence optimal location queries. In min-distance optimal location queries[20], the key objective is to position the facility in a place that minimises the average distance between demand points and facilities based in the new location. Max-influence optimal queries[18] serve to identity a facility location that has maximum attraction to demand points[21] or the total weight of demand points[17] over existing facilities.

Huang et al.[21] formulate the top most influential locations with three components: demand points, existing facilities, and candidate locations. This problem identifies candidate locations that attract the highest number of demand points than the existing facilities, known as the "influence" of candidate locations. It then sorts the candidate locations and selects the first t as a query result. The authors propose two different algorithms for pruning the search space. Similar algorithms are proposed to reduce the search space[17] based only on the influence of existing facilities without considering candidate locations.

Lee et al.[22] extend BRNN and consider the direction as a constraint for a novel type of query called direction-aware bichromatic reverse k nearest neighbour (DBRkNN) queries. The authors provide two efficient algorithms to improve indexing and query processing time. Liu et al.[13] introduce new problems as a KNN-based optimal location query to consider the different labels of facility and probabilities of visiting and relocation KOLQ (RKOLQ) for relocating current facilities to new optimal sites. They suggest two algorithms to address these problems and introduce some novel pruning methods that rely on the k nearest location component (KNLC).

Yilmaz et al.[19] studied the problem of finding a new facility with unknown customer locations. The predictor model consists of two steps, generating customer locations and processing location queries. The first step needs information, such as the number of demand points attracted by existing facilities and auxiliary information. During the generation steps, the model considered the demand points of neighbours, and a Voronoi region for each facility is divided into smaller regions, generating the demand point locations in these regions. The second step in the predictor model involves running the query processor to identify the optimum location for a facility

to attract the maximum number of demand points and minimising the average distance between the demand point and the facility.

To estimate the number of demand points attracted by a given facility, premised on its capacity with 3, 4 and 11 class labels, Zaheer et al.[14] propose a solution for a multi-classes classification problem. Having produced estimates of the customer's location using[19], this model uses a range of features, for example, local information pertaining to schools, and a novel feature, such as the influence of sites. This model was implemented on a range of potential sites in the Punjab region's 36 respective districts comprising 47,926 schools. Following training on existing sites, the model attained an 82% accuracy level for class 3 labels.

Zhdanov et al.[23] propose a solution based on estimating restaurant success data and machine learning approaches for identifying restaurant locations. The authors used data from 2,869 restaurants that included different features, such as geographical and restaurant information and popular attractions. These features were used to train the model to predict success indicators, such as the total number of reviews. The results for the total number of reviews indicated that Random Forest performed better than other models, including mean, random, or CatBoost.

Facility location problems

There are two main categories of facility location problems: median-based and covering-based[3]. Median-based problems are designed to minimise cost to the demand points, median-based problems by analysing potential facility locations and determining the weighted average of cost to these demand points. Fixed-charge facility location problems (FCLPs) and p-median location problems (PMLPs) are examples of median-based problems. FCLPs aim to minimise both travel cost and facility setup costs, while PMLPs focus only on minimising travel costs.

In covering-based problems, facilities are positioned to ensure that demand points fall within a specified travel time or distance range[3,24]. P-centre location problems (PCLPs), maximal covering location problems (MCLPs) and set covering location problems (LSCPs) are considered to be covering-based problems. PCLPs aim to minimise the maximum distance between demand points and facilities, whereas MCLPs aim to maximise the coverage of demand points within a particular travel time or distance range. Similarly, LSCPs seek to minimise the number of facilities needed within a specified travel time or distance range. Conditional FLPs are an extension of FLPs, where new facilities are added, and existing facilities remain[25–29]. The conditional p-median problem locates new facilities in a way that minimises the total weighted distance between demand points and both new and existing facilities[30]. The conditional p-center problem focuses on minimising the maximum distance between new and existing facilities[26,31].

Risanger et al.[32] used pharmacies to select testing sites, aiming to maximise coverage and accessibility to COVID-19 testing for the US population. The researchers drew from a formerly developed model built to achieve the most effective and efficient distribution of drugs during the H1N1 breakout. The inputs were ZIP Code Tabulation Areas (ZCTAs) and the portion of the population willing to travel to testing sites[33]. Data pertaining to population size and centroid coordinates (latitude and longitude) were captured at each ZCTA. 94% of the total population was found to have ready access to pharmacies, with the pharmacies able to serve over 90% of the population in 12 states and over 80% in 47 states.

Akwafuo et al.[34] propose k-means++ clustering to locate PODs. This modification considers two initial points instead of initial random points. The planner selects the number of PODs and then evaluates them using silhouette analysis. This study was carried out in 774 areas in Nigeria and 36 states. To locate new police sub-stations in Pakistan, Mir et al.[35] developed a novel k-median model aimed at minimising the distance between police stations and places where a crime was committed. Stations were to be located close to roads to make them accessible. Experiments were run to identify the performance level of the model in a variety of conditions, e.g., during different times of the day and at different times of the year. Based on a comparison with random selection of locations, the model was found to be effective.

A model for optimising the distribution of vaccinations with set age and distance constraints was produced by Emu et al.[36]. The researchers used priority groups and distance to develop a vaccination model called priority. They also used constraint satisfaction programming (CSP) combined with distance-based vaccine distribution (PD-VDM). K-medoids[37] and the silhouette method was applied in the model to select the number and location of vaccine centres. The model was allocated priority groups to vaccine centres to minimise average distance. It was run on data from Chennai in India and randomly generated data. The results based on the synthetic data and actual data indicated a reduction of travel distance of 40% and 70%, respectively, with100% and 90% vaccination distribution for high-priority groups.

Deng et al.[8] propose an approach to minimise the number of emergency medical service (EMS) facilities needed for existing hospital networks. They used a genetic algorithm (GA) to solve the set covering location problems (LSCPs) titled SCP-GA. The results of SCP-GA showed a reduction of travel time by 14.57% for the weighted median. This study also found that, to achieve 90% of population coverage within a 15-minute travel period, an addition of 55 new EMS facilities is needed. Wang et al.[38] propose a model to minimise the time required to construct PODs and to allocate patients to the PODs. The model was applied in Wuhan, China, and relied on forecasting the number of COVID-19-infected persons. The authors propose a serial-number-coded genetic algorithm (SNCGA) to solve the problem.

Kuvvetli[39] proposes a goal programming model with three objectives: to minimise the distance between demand points and test sampling centres, to minimise the number of test sampling centres, and to minimise the distance between test centres and test sampling centres. The model was applied to two cities in Turkey, resulting in a location-allocation map and the required level of inventory quantities. Alghanmi et al.[15] overcome the issues of needing the distance and number of PODs in advance by proposing an algorithm called points of dispensing location allocation (PODsLA). PODsLA is used to find the number and location of PODs based on

the population size factor. It relies on a novel algorithm called population affinity propagation (PAP). The model was applied to synthetic and real data from the US and a case study based in Jeddah, Saudi Arabia during the COVID-19 pandemic. The results indicated that the proposed algorithm performed better than other algorithms in terms of distance and coverage. It generated the closest results to the case study in terms of number and location of PODs. Tasić et al.[40] propose the conditional p-next centre problem, which addresses two challenges: expanding service networks by adding new facilities while retaining existing ones, and (2) ensuring users are redirected to the closest backup centre if a primary centre fails. They solved the problem metaheuristic, that is, by basing it on the Variable Neighbourhood Search (VSN) called Skewed VNS (SVNS).

Research based on the LSQs concept generally focuses on improving the query process, such as[13,17,21] and[22]. However, Yilmaz et al.[19] processed queries for new facilities with unknown customer locations, while[14] and[23] considered different features to identify the number of demand points or locate facilities. Meanwhile, studies based on FLPs focus on improving access to facilities and coverage without introducing new facilities such as[32,35,36] and[39]. Deng et al.[8] propose an approach that would result in adding the minimum number of new facilities to existing ones. This approach, however, requires predefined travel time and coverage. PODsLA[15] overcomes the need to specify distance and number of PODs in advance while still improving access to facilities or coverage without adding new facilities.

This paper introduces IPLA, an algorithm for adding new PODs to existing ones, which differs from PODsLA. IPLA and PODsLA serve different purposes: PODsLA is designed to establish PODs in scenarios where none currently exist within an emergency response plan, whereas IPLA focuses on adding additional points to existing ones. The PODsLA approach uses PAP clustering, which does not take into account the presence of existing PODs that are already providing services. PAP is designed to initiate the plan and add POD locations at the beginning of an emergency plan, which takes demand point locations and population size as input. The demand points are used as candidate locations and the population size is a factor in determining the location and number of PODs. However, population as a sole independent factor cannot help add new PODs, as existing PODs are already in place. Thus, IPLA is based on the proposed clustering influence affinity propagation (IAP) which allows for the inclusion of new locations even if existing PODs already exist. IAP requires the input of current POD locations and their influence on attracting a greater population, with a minimum distance, to determine the number and location of new POD locations. More details about IPLA are provided in the next section.

Influence-based PODs location allocation algorithm

The emergency planner and health authorities need additional PODs to address service area gaps, ensure broader population coverage, and improve accessibility. In order to achieve this, an influence-based POD location allocation (IPLA) algorithm is proposed for designating the new number and location of PODs when there are existing PODs. IPLA consists of two phases, as shown in Fig. 1; in the first phase, a new weight called the *influence* of PODs is used, which refers to the total population that is attracted by existing POD (e.g., POD *i*) based on the minimum distance to these populations than other PODs (e.g., POD *j*). This factor helps locate more PODs near the most attractive PODs to cover high-demand areas. If one were to assume there is a set of population size *WP* for a set of demand points *DP*, and a set of existing PODs *EP*; the *influence* of current POD is computed as shown in 1.

$$inf(ep_i) = \sum_{wp \in WP} wp \mid \forall\, ep_j \in EP,\ i \neq j,\ dist\,(dp, ep_i) < dist\left(dp, \mathrm{ep}_j\right) \tag{1}$$

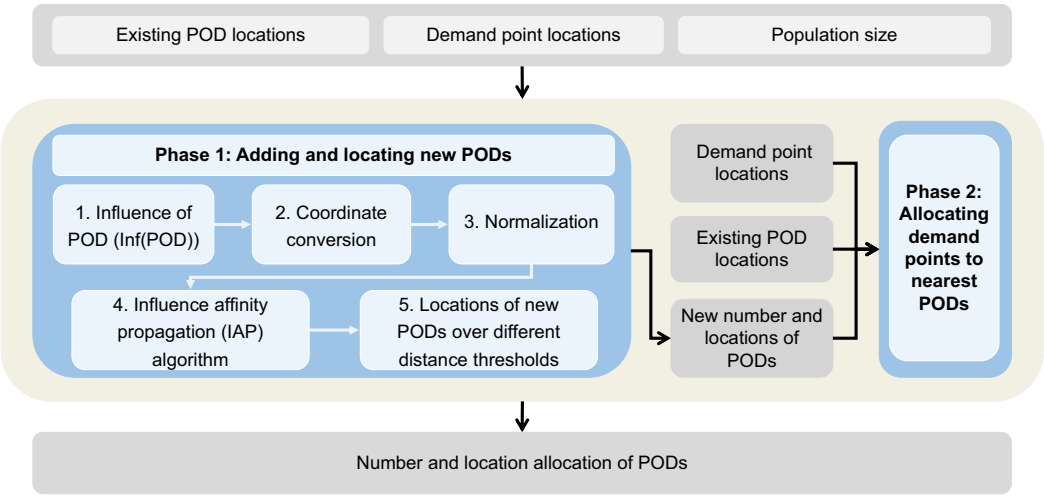


Fig. 1. The workflow of the influence-based algorithm to PODs location allocation (IPLA).

where $inf(ep_i)$ represents the influence of $ep_i \in EP$, the total population size *wp* at *dp* with a minimum distance closer than $ep_j \in EP$. The distance between ep_i, ep_j and *dp* is presented as $dist(dp, ep_i)$ and $dist(dp, ep_j)$, computed using the haversine formula (Equation 2), as road network data was unavailable.

$$\begin{aligned} ah &= \sin^2((\phi_2 - \phi_1)/2) + \cos(\phi_1) \cdot \cos(\phi_2) \cdot \sin^2((\theta_2 - \theta_1)/2) \\ dh &= 2 \cdot \operatorname{atan2}\left(\sqrt{ah}, \sqrt{1-ah}\right) \cdot r \end{aligned} \tag{2}$$

where, respectively, the latitude and longitude of sites 1 and 2 are represented by ϕ_1, ϕ_2, θ_1, and θ_2, and $r = 6371000\ m$ is the earth's mean radius.

Pre-processing steps, namely coordinate conversion and normalisation[15], are then applied in the IPLA approach. POD locations are normalised into [-1,1], and the influence of PODs are normalised into [0,1]. Spherical coordinates of POD locations are transformed into Cartesian coordinates[41]. Then, the proposed influence affinity propagation (IAP) algorithm finds the new number of PODs and identifies the areas where they will be placed. IAP is based on the affinity propagation (AP), which is a clustering algorithm that differs from other K-centres in that no initial number of clusters is required[42,43]. AP assigns equal weight to all demand points, leading to an even distribution of PODs without considering highly populated areas. PAP[15] overcomes this limitation by incorporating population size as a design parameter to locate and determine the number of PODs in highly populated regions. Our IAP considers the influence of existing PODs as a factor in determining the number and location of new PODs.

Preferences and a similarity matrix are inputs for AP, while PAP and IAP incorporate population size and the influence of existing PODs, respectively, into the similarity matrix. Additionally, IAP uses the location of existing PODs as candidate locations for new PODs, rather than using all demand points as in PAP[15]. Consider a dataset $S = (ep_i, inf(ep_i))$ that consists of ep_i which is the location of existing POD and $inf(ep_i)$ which is the total population that is attracted by ep_i (Equation 1). ep_i comprises three attributes $\{X_i, Y_i, Z_i\}$ that constitute the location of ep_i in Cartesian coordinates, and negative Euclidean distance as $dist(ep_i, ep_k)$ is used to discern the distance between ep_i and ep_k. In 3, the definition of the weighted similarity matrix between data points ep_i and ep_k is given.

$$s(i,k) = \begin{cases} -inf(ep_i) \cdot dist(ep_i, ep_k) & \text{if } i \neq k, \\ p & \text{if } i = k, \end{cases} \tag{3}$$

where *p* is set to the similarity matrix's median and the first quantile. The rationale behind selecting these values is that they can be used to find the number of PODs or to reduce that number. It refers to that in reminder as: IPLA and $\text{IPLA}_{q=0.1}$.

The data points exchange two different types of messages in IAP. Responsibility: *r*(*i*, *k*), which is sent from data point i to the candidate exemplar point k. The purpose of this is to assess k's eligibility as an exemplar for i while accounting for potential exemplars (Equation 4)[44]. The second is availability: $a(i,k)$ is sent from the candidate exemplar point k to the point i to assess if i is suitable to choose k as an exemplar. This is possible as other data points support the idea that k should be an exemplar (Equation 5)[44].

$$r(i,k) = s(i,k) - \max_{k':k' \neq k} \left[s\left(i,k'\right) + a\left(i,k'\right) \right] \tag{4}$$

$$a(i,k) = \begin{cases} \sum_{i':i' \neq k} \max\left[0, r\left(i',k\right)\right], & k = i \\ \min\left[0, r(k,k) + \sum\limits_{i':i' \notin \{i,k\}} \max[0, r(i',k)]\right], & k \neq i \end{cases} \tag{5}$$

Lastly, the sum of $a(i,k)$ and $r(i,k)$ is produced as the creation matrix. The highest creation value is chosen as the exemplar in each row (point i), and rows with a common exemplar are clustered together.

$$\hat{c}_i = \operatorname{argmax}_k [a(i,k) + r(i,k)] \tag{6}$$

The centroid of each demand area is used as a candidate location for the existing PODs (one centroid for each area). However, once the number and locations of the new PODs are designated, these existing POD locations cannot be used to compute coverage and distance more accurately. Therefore, IPLA uses Equation 7 to generate new locations at various distances[45,46].

$$\begin{aligned} \phi_n &= \operatorname{asin}\left(\sin\phi_c \cdot \cos\frac{d}{r} + \cos\phi_c \cdot \sin\frac{d}{r} \cdot \cos\lambda\right) \\ \theta_n &= \theta_c + \operatorname{atan}2\left(\sin\lambda \cdot \sin\frac{d}{r} \cdot \cos\phi_c, \cos\frac{d}{r} - \sin\phi_c \cdot \sin\phi_n\right) \end{aligned} \tag{7}$$

where ϕ_n and θ_n are the new latitude and longitude of the existing latitude and longitude of the location of the POD ϕ_c and θ_c. λ is the bearing (clockwise from north), d the required distance to be travelled, and d/r is the angular distance. d is set over different distance thresholds, which means that the new POD locations range from 1 to 5 miles away from the existing POD locations. To make this easy to follow, we denote it in the reminder as l1 to l5.

Finally, the number and locations of all new and existing PODs are merged in order to allocate the demand points to the closest POD using the haversine formula (Equation 2). We follow the assumption that people located in the same area would frequent the POD located in that area[32,33]. Thus, the distance between the POD and the residents of the same demand point is calculated as the square root of the demand area divided by two for the full population.

Algorithm 1 presents the IPLA algorithmically, using population size, demand point locations, and the locations of existing PODs to determine the location and number of new PODs. The demand points are then allocated to the PODs that are nearest to them based on the final number and location of PODs.

Input: DP: demand points locations, EP: existing PODs locations, and WP: population size
Output: N: The number of PODs, LP: location of PODs, and allocation to PODs
Calculate the influence of existing PODs as $inf(EP)$ ▷ Equation 1
Represent the coordinates of EP in Cartesian coordinates as EP_c
Normalize EP_c and $inf(EP)$ into [-1,1] and [0,1]
Employ IAP $(EP_c, inf(EP))$ algorithm to identify new PODs as NP
Generate new locations NP_g based on different distance thresholds ▷ Equation 7
Merge EP and the new PODs NP_g as LP
Find the total number of PODs LP as N
Allocate the demand points DP to nearest LP ▷ Equation 2

Algorithm 1. Influence-Based for PODs Location-Allocation Algorithm.

Experiment setting

In order to determine the number and location of existing PODs, the PODsLA algorithm is used[15] to demand point locations and population size, which are needed as input by SCP-GA[8], TILS[14], IPLA, and $\text{IPLA}_{q=0.1}$ to get new PODs. PODsLA, TILS, and SCP-GA paradigms represent diverse methodologies, including clustering-based, query-based, and optimisation-based techniques. These comparisons effectively demonstrate the strengths and novelty of our approach. The PODsLA approach enables the evaluation of improvement in coverage and distance compared to other approaches. In real-world scenarios, locations of existing PODs are typically determined by emergency planners based on their current response plan. The aggregation of both already-present PODs and new PODs comprises the final number and location of PODs.

IPLA and PODsLA[15] algorithms require two pre-processing steps – normalisation and coordinate conversion. Latitude and longitude coordinate values are made into Cartesian coordinates from spherical ones, because these match the locations on the surface of the Earth, which is considered a sphere in non-Euclidean space. However, the Euclidean distance measures the distance in Euclidean space between two points[47], where it is needed for clustering algorithms in IPLA and PODsLA[15]. Normalisation is a useful step for methods that use distance measurements as clustering algorithms[48].

TILS finds the influence of sites based on the total population attracted by existing PODs, which is based on a query-based approach. Furthermore, as IPLA, Equation 7 is applied in TILS to generate new locations at various distances to more accurately compute the coverage and distance. TILS then needs the t in advance, where the minimum t is selected based on a range [3-10] when the values of the population coverage do not change more than two times. The SCP-GA, which is based on optimisation methodologies, uses existing PODs to get the current population coverage, the desired percentage by IPLA to get the percentage gap, and the predetermined travel distance threshold selected from 1 to 5 miles. For example, the existing PODs cover 72.5% of the population within a distance threshold of 3 miles; thus, to reach 80%, the SCP-GA needs the percentage gap between 72.5% and 80%, which is 7.5%, besides the existing PODs and travel distance threshold. SCP-GA, TILS, IPLA, and $\text{IPLA}_{q=0.1}$ are compared to one another and to PODsLA.

Data and study area

Four datasets are used in this study, three of which are synthetic datasets, and one which is a real dataset taken from the Jeddah case study dataset compiled during the COVID-19 pandemic. Demand point locations, presented as longitude and latitude, and population size, are selected from each dataset. Below is a brief description of each dataset:

- **Synthetic datasets:** There are three states within this dataset, which serves as a fictitious representation of the areas and pharmacy databases[32,49]. Each state has different demand points as 501, 201 and 301 for ONE, TWO and THREE states, receptively. Also, the total population for each is 5,027,539, 3,954,307 and 1,463,970, respectively. This dataset includes attributes such as the land area and population size with their locations based on ZCTAs. These locations are represented as longitude and latitude for the centroid of each ZCTA. The ZIP Code service areas served by the United States Postal Service (USPS) are represented generally by ZCTAs[50]. According to the USPS, ZIP codes are a grouping of postal delivery routes rather than topographical features.
- **Jeddah dataset:** The total area of Jeddah is approximately 1765 km^2, with a total population of 3,628,129. Jeddah's three main healthcare providers are the Ministry of Health's network of hospitals and primary care clinics (spread throughout the city), other governmental organisations, and the private sector[51]. This study

covers only medical facilities that are managed by the Ministry of Health. Various reliable sources are used to obtain information on the districts, population size, and COVID-19 vaccination centres. The information on distribution within the 156 districts is collected from the municipality of Jeddah[52], while the General Authority of Statistics provided the population data[53], which were then manually inputted for each district. When collecting population data, we consider the entire population, without differentiating or focusing on specific vulnerable groups. Note that the population data is collected as census data and does not reflect any individual or contain private information. The list of vaccine centres was provided by the MOH[6]. There were 12 vaccination centres in 2020, which increased by 30 to 42 centres by 2021. The 42 PODs are all included in this study, as shown in Fig. 2.

Performance evaluation

R language is used in our experiment to implement the algorithm and generate results, with the code publicly available on GitHub (https://github.com/nrghanmi/IPLA.). The proposed algorithm is evaluated and assessed on the basis of different evaluation metrics: population coverage, weighted distance, and prediction measures. Population coverage is calculated as the sum of population size at demand points according to the distance threshold (with a range of 1-10 miles). The following shows the calculation of the average weighted distance of the population to access the PODs:

$$\bar{x} = \frac{\sum_{i=1}^{DP} wp_i \cdot x_i}{\sum_{i=1}^{DP} wp_i} \tag{8}$$

where *wp* is the population size at demand point *i*, *x* is the distance for demand points to access nearest PODs.

The PODs predicted by the proposed algorithm are evaluated against the actual POD locations (Jeddah dataset) based on prediction measures, namely, true positive rate (TPR), false positive rate (FPR), F_1 score, and area under the receiver operating characteristic (ROC) curve (AUC). The closest PODs (predicted and actual) across distance thresholds falling within the range of 1 and 5 miles are used to calculate these metrics. By calculating the distances between predicted and actual POD locations and considering only those locations that fit the required distance threshold, the closest locations can be found. The minimum distance, where there is no prospective best matching that has the minimum distance or where there is only one POD within the requisite distance thresholds, forms the basis for the final best matching.

TPR or recall measures how predicted POD locations are nearest to actual POD locations within the 1-5 miles range. A high score of TPR is required, which means that predicted and actual POD locations are near and matched within the required threshold. FPR measures how predicted POD locations are not near or matched within 1-5 miles to actual POD locations. A low score of FPR is required, which means that within the required threshold, a few predicted POD locations are not near to the actual locations.

F_1 score combines the recall and precision to find the harmonic mean among them, where precision refers to corrected predicted POD locations out of all predictions. AUC measures a two-dimensional area under the

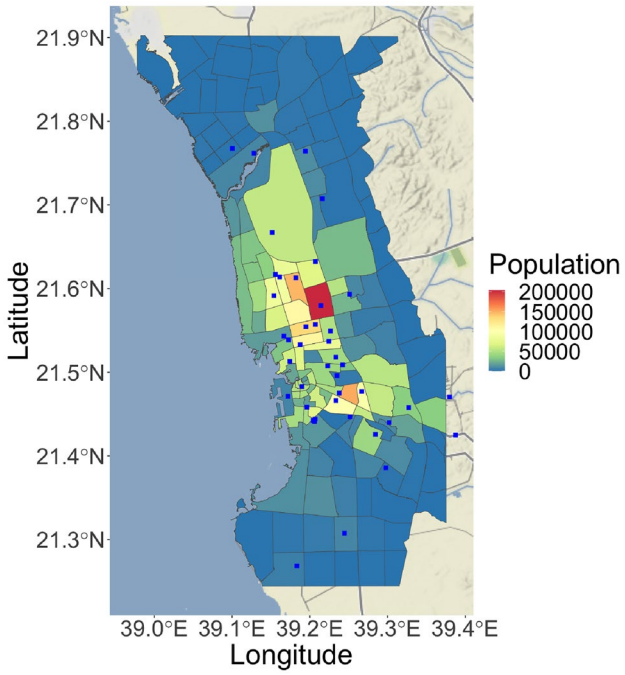


Fig. 2. Population distribution and POD locations in districts of Jeddah. Map is created using ggmap[54] package in R version 4.4.2 and map from Stadia Maps[55].

Evaluation metric	Formula
True Positive Rate (TPR)/Recall	$\frac{TP}{\text{Total number of } POD_A}$ $TP = \lvert\{d(pa, pd) \leq d\}\rvert$
False Positive Rate (FPR)	$\frac{FP}{\text{Total number of } POD_P}$ $FP = \lvert\{d(pa, pd) > d\}\rvert$
Precision	$\frac{TP}{TP+FP}$
F_1 score	$2 \cdot \frac{(Recall \cdot Precision)}{(Recall + Precision)}$

Table 1. Prediction performance metrics.

PODsLA	TILS (t=+8)	SCP-GA (np_n=+{2,3,4})	$IPLA_{q=0.1}$ (np_n=+5)	IPLA (np_n=+9)
43	51	{45,46,47}	48	52

Table 2. The total number of PODs for ONE state using the PODsLA algorithm or by summing the PODsLA algorithm with the TILS, SCP-GA, IPLA, and $IPLA_{q=0.1}$ approaches. Where t refers to the input parameter for new PODs used in TILS and np_n represents the number of new PODs produced by SCP-GA, IPLA, and $IPLA_{q=0.1}$.

Distance threshold (in mile)	PODsLA (np=43)	TILS (np=51)	$IPLA_{q=0.1}$ (np=48)	IPLA (np=52)
1	14.426	15.421	15.037	15.421
2	23.462	26.267	25.316	26.635
3	36.104	40.054	38.622	40.311
4	53.014	56.616	56.352	57.128
5	66.690	69.69	69.618	70.171
6	76.715	79.377	79.334	80.079
7	88.632	90.388	89.955	91.3
8	95.995	97.117	96.875	98.057
9	98.578	99.109	98.943	99.169
10	99.506	99.506	99.506	99.506

Table 3. The population coverage over a 1–10-mile distance for PODsLA, TILS, IPLA, and $IPLA_{q=0.1}$ approaches – ONE state. Where np refers to the total number of PODs. IPLA, $IPLA_{q=0.1}$, and TILS locate new PODs 5 miles from the PODs produced by PODsLA.

probability curve ROC, where the FPR and TPR are on the x-axis and y-axis, respectively. AUC measures the trade-off between FPR and TPR within a range of values [0,1]. A higher value of the AUC score means the predictive power is good in distinguishing between actual and predicted locations. In other words, the algorithm has a higher TPR without a significant increase in FPR.

Table 1 summarises the formula of the described prediction metrics. In Table 1, true positive (TP) denotes actual POD locations $pa \in POD_A$ that are correctly identified within the distance threshold d as near to predicted POD locations $pd \in POD_P$. False positive (FP) represents the number of predicted POD locations $pd \in POD_P$ that are not near actual POD locations $pa \in POD_A$ within the distance threshold d.

Results and discussion

The total number of PODs of state ONE for PODsLA[15], SCP-GA[8], TILS[14], IPLA, and $IPLA_{q=0.1}$ are presented in Table 2. In ONE state, there is improvement in population coverage, reaching more than 3% and 4%, while in the average weighted distance, the enhancement reaches around 0.3 and 0.4 kilometers, as shown in Tables 3 and 4. $IPLA_{q=0.1}$ shows a similar population coverage and average weighted distance to other algorithms but a low number of PODs with 5 new locations, while there are some exceptions within 2 and 3 miles which has a lower population coverage than others. The IPLA algorithm produces more PODs than other algorithms, with 9 new locations, and TILS needs as input of 8 new locations.

In TWO state, there is improvement in population coverage, as shown in Table 5, with 7%, while in the average weighted distance, the enhancement reaches around 0.3 and 0.4 kilometers. The proposed algorithms produce fewer new locations, with 6 and 4, while the current work needs 7. IPLA and TILS yield similar results

Predetermined distance threshold (in mile)	PODsLA (np=43)	SCP-GA (np={45,46,47})	TILS (np=51)	$\mathbf{IPLA}_{q=0.1}$ (np=48)	IPLA (np=52)
1	6,978.212	6,927.407	6,655.663	6,710.247	6,577.942
2		6,802.773			
3		6,754.574			
4		6,765.865			
5		6,774.061			

Table 4. The average weighted distance in metres over a predetermined distance from 1–5 miles for the SCP-GA algorithm against other algorithms – ONE state. Where np refers to the number of PODs. IPLA, $\mathrm{IPLA}_{q=0.1}$, and TILS locate new PODs 5 miles from the PODs produced by PODsLA.

Distance threshold (in mile)	PODsLA (np=23)	TILS (np=30)	$\mathbf{IPLA}_{q=0.1}$ (np=27)	IPLA (np=29)
1	19.871	20.792	20.752	20.792
2	29.547	32.865	32.28	33.388
3	42.029	49.536	47.315	49.419
4	59.131	66.189	62.752	66.16
5	70.556	77.002	73.397	74.999
6	84.424	90.099	87.382	87.535
7	92.152	93.955	92.445	92.445
8	95.663	96.579	95.663	95.663
9	98.133	98.133	98.133	98.133
10	99.164	99.831	99.164	99.164

Table 5. The population coverage over a 1–10-mile distance for PODsLA, TILS, IPLA, and $\mathrm{IPLA}_{q=0.1}$ approaches – TWO state. Where np refers to the total number of PODs. IPLA, $\mathrm{IPLA}_{q=0.1}$, and TILS locate new PODs 5 miles from the PODs produced by PODsLA.

Actual PODs (Jeddah COVID-19 vaccination centers)	PODsLA	TILS (t=+3)	SCP-GA (np_n=+1)	$\mathbf{IPLA}_{q=0.1}$ (np_n=+4)	IPLA (np_n=+6)
42	35	38	36	39	41

Table 6. The total number of actual PODs in Jeddah city against PODsLA, TILS, SCP-GA, $\mathrm{IPLA}_{q=0.1}$, and IPLA. Where t refers to the input parameter for new PODs used in TILS and np_n represents the number of new PODs produced by SCP-GA, IPLA, and $\mathrm{IPLA}_{q=0.1}$.

for population coverage, with some exceptions, such as within the 5 and 6 mile mark. In THREE state, the number of new PODs for the proposed algorithms is fewer than current work. The population coverage for $\mathrm{IPLA}_{q=0.1}$ is improved by around 2% and around 5% for other algorithms. The average weighted distance is improved to reach around 0.4 or 0.5 kilometers for IPLA and TILS.

In the Jeddah dataset, the total number of PODs using the proposed algorithms are close to the actual PODs, as shown in Table 6. The algorithms show a slight difference in population coverage and average weighted distance compared to PODsLA, as shown in Tables S.18 and S.19 in the supplementary material. Thus, the predictive performance for the Jeddah dataset is computed to compare the predicted PODs with actual PODs. In TPR, both proposed algorithms show the highest scores, which means that many predicted PODs are nearest to actual within 1 to 5 miles, as shown in Table 7. However, the current algorithms have the lowest scores in FPR, meaning that a few predicted PODs are not near to actual within the 1 to 5 mile range. More importantly, in Table 8, $\mathrm{IPLA}_{q=0.1}$ has over 0.7 in AUC four times when the new POD locations range 2 to 5 miles away from the existing POD locations. Meanwhile, TILS has reached over 0.7 twice, however, it has the highest score in F_1 compared with all other algorithms, as shown in Table 9.

The SCP-GA uses the PODs from PODsLA and the percentage gap to reach the desired population coverage, which is set as proposed by the IPLA. These parameters will lead to a different number of new PODs. The distance and the gap coverage play a role in determining the number of new PODs; while 4 PODs are needed within 2 miles to reach 3% of population coverage, 3 are needed within 5 miles in ONE state. The same thing is applied for TWO state; it produces 2 or 3 to reach 4% of population coverage within 2 and 5 miles, respectively. Tables S.1-S.5 in the supplementary material show the number of PODs, average weighted distance and population coverage pertaining to other states. The results regarding population coverage, when the new PODs for the algorithms are located far from the PODs determined by PODsLA (within 1 to 4 miles), can be found in Tables

Distance threshold for nearest location	Approaches	l1	l2	l3	l4	l5
1	PODsLA (np=35)	0.429				
	SCP-GA (np=36)	0.452				
	TILS (np=38)	**0.5**	0.476	0.452	0.429	0.452
	IPLA (np=41)	**0.5**	**0.5**	0.452	0.452	0.452
	$\mathrm{IPLA}_{q=0.1}$ (np=39)	**0.5**	**0.5**	0.476	0.452	0.476
2	PODsLA (np=35)	0.643				
	SCP-GA (np=36)	0.667				
	TILS (np=38)	0.69	0.667	0.667	0.69	0.667
	IPLA (np=41)	0.69	0.69	**0.714**	0.667	0.69
	$\mathrm{IPLA}_{q=0.1}$ (np=39)	0.69	0.643	0.667	0.667	0.667
3	PODsLA (np=35)	0.667				
	SCP-GA (np=36)	0.69				
	TILS (np=38)	0.714	0.714	0.738	0.714	0.714
	IPLA (np=41)	0.738	**0.762**	0.738	0.738	**0.762**
	$\mathrm{IPLA}_{q=0.1}$ (np=39)	0.714	0.738	0.714	0.738	0.714
4	PODsLA (np=35)	0.762				
	SCP-GA (np=36)	0.762				
	TILS (np=38)	0.81	**0.833**	**0.833**	**0.833**	**0.833**
	IPLA (np=41)	**0.833**	**0.833**	**0.833**	**0.833**	**0.833**
	$\mathrm{IPLA}_{q=0.1}$ (np=39)	0.81	0.81	**0.833**	**0.833**	**0.833**
5	PODsLA (np=35)	0.786				
	SCP-GA (np=36)	0.81				
	TILS (np=38)	0.833	0.833	0.833	0.857	0.857
	IPLA (np=41)	0.857	0.857	0.857	0.857	0.857
	$\mathrm{IPLA}_{q=0.1}$ (np=39)	0.833	0.857	**0.881**	**0.881**	**0.881**

Table 7. The TPR for PODsLA, SCP-GA, TILS, IPLA, and $\mathrm{IPLA}_{q=0.1}$ over distances ranging from 1–5 miles. l1, ..., and l5 represent the locations of new PODs located far (1–5 miles) from the PODs produced by PODsLA. Where np refers to the total number of PODs.

Approaches	l1	l2	l3	l4	l5
PODsLA (np=35)	0.686				
SCP-GA (np=36)	0.698				
TILS (np=38)	**0.711**	0.699	0.687	0.688	**0.7**
IPLA (np=41)	0.689	0.689	0.665	0.665	0.665
$\mathrm{IPLA}_{q=0.1}$ (np=39)	0.699	**0.712**	**0.712**	**0.701**	**0.712**

Table 8. The AUC for PODsLA, SCP-GA, TILS, IPLA, and $\mathrm{IPLA}_{q=0.1}$. l1, ..., and l5 represent the locations of new PODs located far (1–5 miles) from the PODs produced by PODsLA. Where np refers to the total number of PODs.

S.6-S.17. Additionally, Table S.20 displays the results for FPR, while Tables S.21, S.22, and S.23 present the results for SCP-GA.

TILS shows some fluctuation in the new number of PODs; it shows a higher number of PODs, such as in synthetic datasets, compared with the proposed algorithms but without significant improvement in coverage and distance. Therefore, competition remains between the two proposed solutions; the decision depends on coverage and distance. For example, in ONE state, the improvement reached around 1.5% in population coverage when more PODs, for example 9 PODs, were added. This implies that it would perform better when selecting 5 new PODs. However, looking at other states, adding more PODs increased coverage to 1.5% - 3.5%. In the Jeddah dataset, before measuring the prediction performance, selecting 4 PODs is better than selecting 6 PODs since there is no improvement in coverage and distance. This point has been validated using prediction measurement.

The distribution of PODs using PODsLA, combined with new PODs generated by different algorithms for the Jeddah dataset, is illustrated in Fig. 3. The SCP-GA (Fig. 3b) places a new POD in a highly populated area; however, several other areas remain underserved. In contrast, TILS (Figure 3c) places two new PODs in close areas, which still leaves coverage gaps in other areas. On the other hand, IPLA (Figure 3d) distributes most of the new PODs in previously uncovered areas, thereby improving overall coverage. Additionally, $\mathrm{IPLA}_{q=0.1}$ (Fig.

Distance threshold for nearest location	Approaches	l1	l2	l3	l4	l5
1	PODsLA (np=35)	0.468				
	SCP-GA (np=36)	0.487				
	TILS (np=38)	**0.525**	0.5	0.475	0.45	0.475
	IPLA (np=41)	0.506	0.506	0.458	0.458	0.458
	$\mathrm{IPLA}_{q=0.1}$ (np=39)	0.519	0.519	0.494	0.469	0.494
2	PODsLA (np=35)	0.701				
	SCP-GA (np=36)	0.718				
	TILS (np=38)	**0.725**	0.7	0.7	**0.725**	0.7
	IPLA (np=41)	0.699	0.699	0.723	0.675	0.699
	$\mathrm{IPLA}_{q=0.1}$ (np=39)	0.716	0.667	0.691	0.691	0.691
3	PODsLA (np=35)	0.727				
	SCP-GA (np=36)	0.744				
	TILS (np=38)	0.75	0.75	**0.775**	0.75	0.75
	IPLA (np=41)	0.747	0.771	0.747	0.747	0.771
	$\mathrm{IPLA}_{q=0.1}$ (np=39)	0.741	0.765	0.741	0.765	0.741
4	PODsLA (np=35)	0.831				
	SCP-GA (np=36)	0.821				
	TILS (np=38)	0.85	**0.875**	**0.875**	**0.875**	**0.875**
	IPLA (np=41)	0.843	0.843	0.843	0.843	0.843
	$\mathrm{IPLA}_{q=0.1}$ (np=39)	0.84	0.84	0.864	0.864	0.864
5	PODsLA (np=35)	0.857				
	SCP-GA (np=36)	0.872				
	TILS (np=38)	0.875	0.875	0.875	0.9	0.9
	IPLA (np=41)	0.867	0.867	0.867	0.867	0.867
	$\mathrm{IPLA}_{q=0.1}$ (np=39)	0.864	0.889	**0.914**	**0.914**	**0.914**

Table 9. The F_1 for PODsLA, SCP-GA, TILS, IPLA, and $\mathrm{IPLA}_{q=0.1}$ over distances ranging from 1–5 miles. l1, ..., and l5 represent the locations of new PODs located far (1–5 miles) from the PODs produced by PODsLA. Where np refers to the total number of PODs.

3e) allocates all new PODs across four distinct areas, targeting both less densely and highly populated areas to better address coverage needs.

The agreement between the predicted and actual PODs within a one-mile radius is analysed by comparing the highest and lowest number of new PODs matched with actual PODs, as shown in Fig. 4 and Figure S.1 in the supplementary material. This analysis considers cases where the new PODs are located one and four miles away from the existing PODs. The proposed algorithm consistently matches one to three new PODs with actual PODs, achieving the highest agreement across these scenarios. In contrast, the TILS approach shows either no matches or, at most, three matches. These results demonstrate the effectiveness of the proposed algorithm in aligning new POD placements with the actual location of PODs in the Jeddah case study.

SCP-GA has the highest execution time due to the need to search through all demand points to optimise the coverage objective. On the other hand, TILS has the lowest execution time, which is nearly negligible and significantly faster than other algorithms. For PAP and IAP, all methods are executed ten times, and the average execution time is recorded. For PAP, in the ONE, THREE, and TWO states, the execution times are 1.099, 0.27, and 0.109 seconds, respectively. This can be attributed to the need to search all demand points to determine the location of PODs. In contrast, for IAP, the execution time is consistently less than 0.001 seconds across all states. This is expected, as IAP does not require searching through all demand points to locate new PODs; instead, it relies on the existing POD locations to identify new positions for PODs.

Ablation study and analysis

The weight of similarity matrix in IAP is calculated on the basis of the mean and maximum of influence weight between two locations as shown in (9) and (10). These are called the IPLA-Mean and IPLA-Max approaches. The number of PODs should be fewer, but should positively affect distance and coverage. However, using the maximum and mean between two POD locations weight does not produce better results. The reason for this is that using maximum or mean values leads to a symmetric matrix, which assigns equal importance to all locations. This uniform weighting fails to account for varying levels of population attraction at different locations. On the other hand, using only weight i, as defined in Equation 3 for the IAP algorithm, results in an asymmetric matrix. This approach assigns higher importance to locations that attract a bigger population, effectively prioritising these locations over others. This weighting strategy better reflects the real-world dynamics of population distribution and improves the overall coverage.

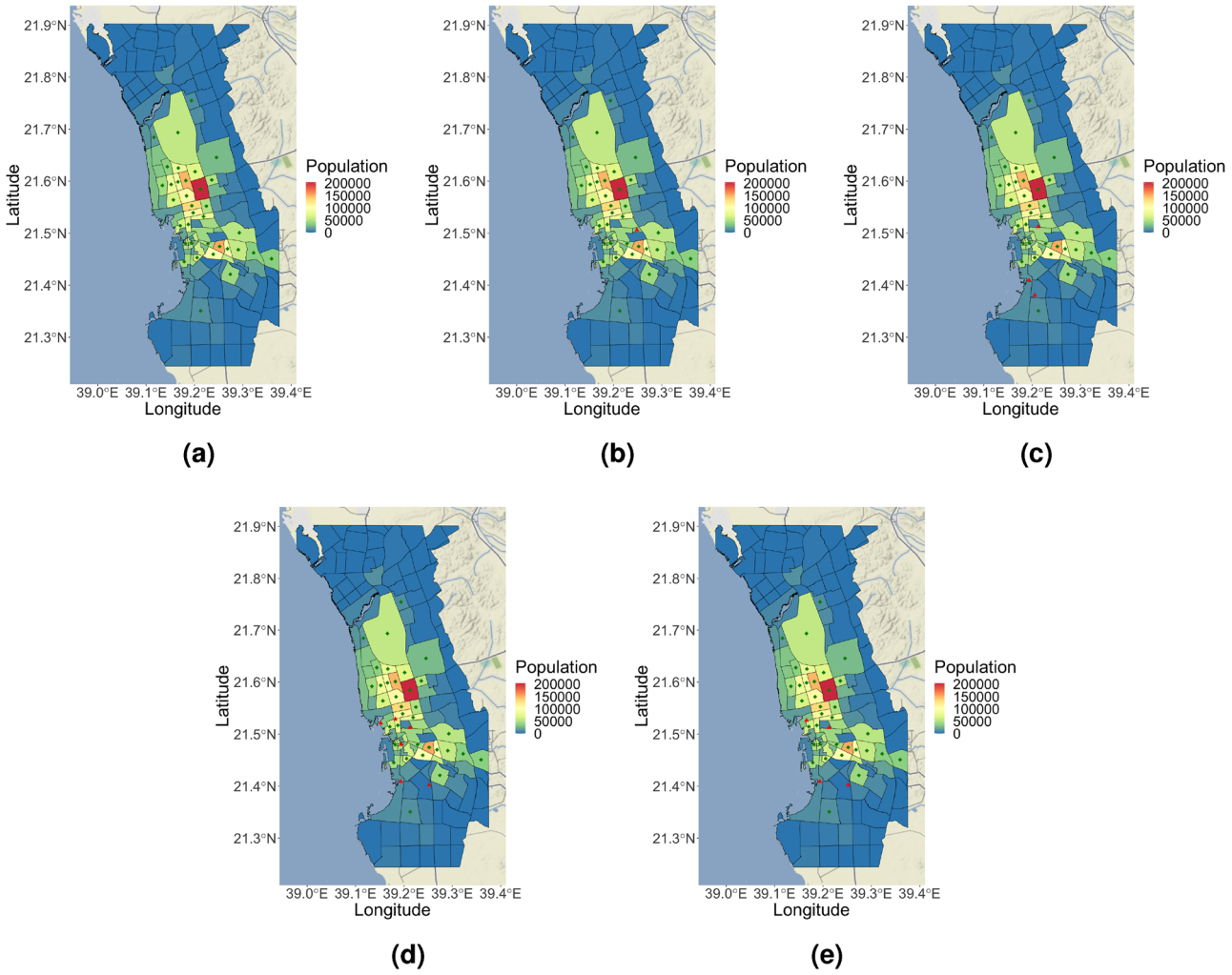


Fig. 3. The distribution of PODs by (**a**) PODsLA, (**b**) SCP-GA, (**c**) TILS, (**d**) IPLA, and (**e**) $\mathrm{IPLA}_{q=0.1}$. The small red triangle represents the new PODs, while the small green diamond represents the existing PODs, for PODsLA. Maps are created using ggmap[54] package in R version 4.4.2 and map from Stadia Maps[55].

$$s(i,k) = -\text{mean}(inf(ep_i), inf(ep_k)) \cdot dist(ep_i, ep_k)), i \neq k \tag{9}$$

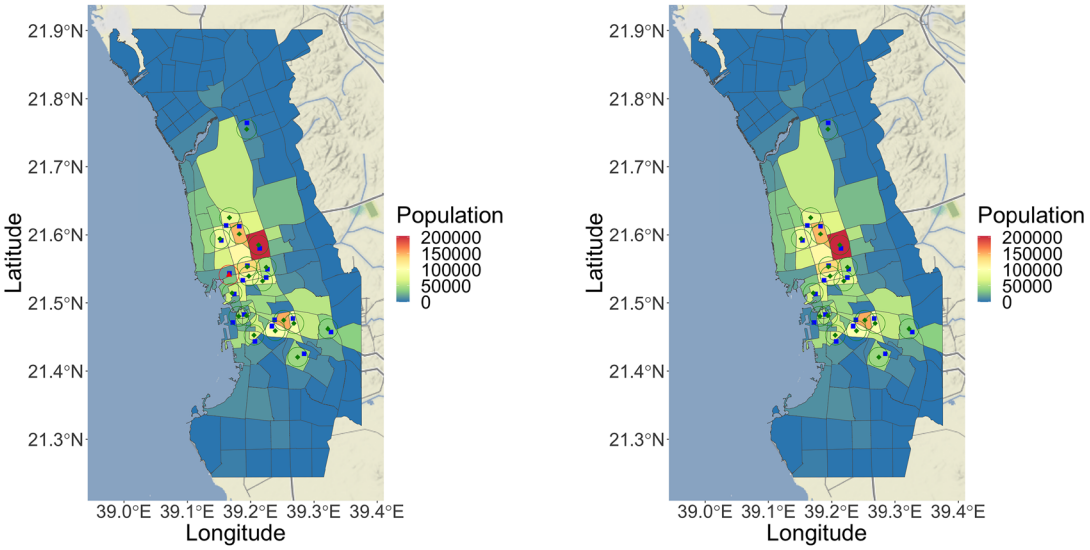
$$s(i,k) = -\max(inf(ep_i), inf(ep_k)) \cdot dist(ep_i, ep_k)), i \neq k \tag{10}$$

IPLA-Mean and IPLA-Max approaches resulted in fewer PODs, which negatively impacted coverage and distance compared to our proposed IPLA. For example, as shown in Table 10 when the new PODs are located 5 miles from the current PODs, using the IPLA-Mean and IPLA-Max improved coverage of existing PODs over a distance of 3 miles by approximately 1%, but the IPLA approach improved the coverage by around 4%. The average weighted distance did not perform well using IPLA-Mean and IPLA-Max, as the level of improvement reached 0.2 kilometers or less. In contrast to the IPLA approach, the distance improvement reached approximately 0.4 kilometers. In the supplementary material, Table S.24 shows the results in terms of average weighted distance. Table S.25 shows the results in terms of population coverage.

At the same time, IPLA-Max and IPLA-Mean locate new PODs in areas with lower population size (box numbers 1–3 in Fig. 5a and Figure S.2 in the supplementary material), which does not guarantee the coverage of more population. However, Figure 5b highlights areas with high demand that were previously underserved by existing PODs. To address this, IPLA designates new PODs that either increase population coverage (box numbers 1–3) or reduce travel distance for highly populated areas to access the nearest POD (box numbers 4 and 5). This strategy demonstrates IPLA's ability to cover a bigger population size and efficiency in POD distribution, with Fig. 5a and b showing how the algorithm addresses both low and high population regions.

Conclusion

During public health emergencies, such as infectious disease outbreaks, a response plan is initiated that involves establishing PODs to provide health services to the affected population. Once the emergency response plan is



(a) IPLA$_{q=0.1}$, the minimum number of predicted locations matching actual POD locations.

(b) TILS, no new predicted locations are matched to actual POD locations.

Fig. 4. The predicted and actual POD locations that are nearest to each other, within a mile (circle frames), for our proposed IPLA$_{q=0.1}$ and the TILS approaches - Jeddah. The new PODs are located 4 miles from the existing PODs, for PODsLA. The small red triangle and the green diamond represent the new PODs and existing PODs by PODsLA, while the small blue square represents the actual PODs. Maps are created using ggmap[54] package in R version 4.4.2 and map from Stadia Maps[55].

	IPLA np=52		IPLA-Mean np=49		IPLA-Max np=49	
Distance threshold (in mile)	l1	l5	l1	l5	l1	l5
1	14.734	15.421	14.613	14.426	14.575	14.426
2	24.245	26.635	24.413	23.462	23.686	23.462
3	38.3	40.311	37.002	37.25	36.842	37.088
4	54.138	57.128	53.195	55.3	53.515	55.517
5	67.819	70.171	67.258	69.654	67.07	69.786
6	77.843	80.079	77.126	78.563	77.141	79.752
7	89.473	91.3	88.926	89.326	89.299	90.107
8	96.989	98.057	95.995	96.217	96.16	96.383
9	98.638	99.169	98.578	98.778	98.578	98.779
10	99.506	99.506	99.506	99.506	99.506	99.506

Table 10. The population coverage over a 1–10-mile distance for IPLA, IPLA-Mean, and IPLA-Max approaches – ONE state. l1, ..., and l5 represent the locations of new PODs located far (1–5 miles) from the PODs produced by PODsLA. Where np refers to the total number of PODs.

initiated, it may be necessary to improve on it by adding and locating more PODs. This paper proposed the IPLA algorithm to locate more PODs to supplement existing PODs. IPLA is based on the novel weight called 'influence of existing PODs' and the IAP clustering algorithm. The influence of existing PODs refers to the total population size attracted by existing PODs with minimum distance. Our proposed IAP algorithm uses this novel weight with geographical information to designate the number of new PODs and their location.

The results of this study show an improvement in population coverage with a lower number of PODs compared to other approaches, such as query-based approach. In terms of the case study of the city of Jeddah, IPLA produced good results using the prediction measures TPR and AUC when compared to other approaches, and yielded comparable results in F_1 score. More importantly, IPLA distributes new PODs in more uncovered areas than other approaches. Notably, this study demonstrated that using the mean or maximum of two locations as a weight for designating the new number and locations of PODs may not be helpful to either reduce the number of PODs or increase the size of the population covered. The influence-based approach is particularly useful in areas where improving coverage and improving access is a priority, for example where the aim is to

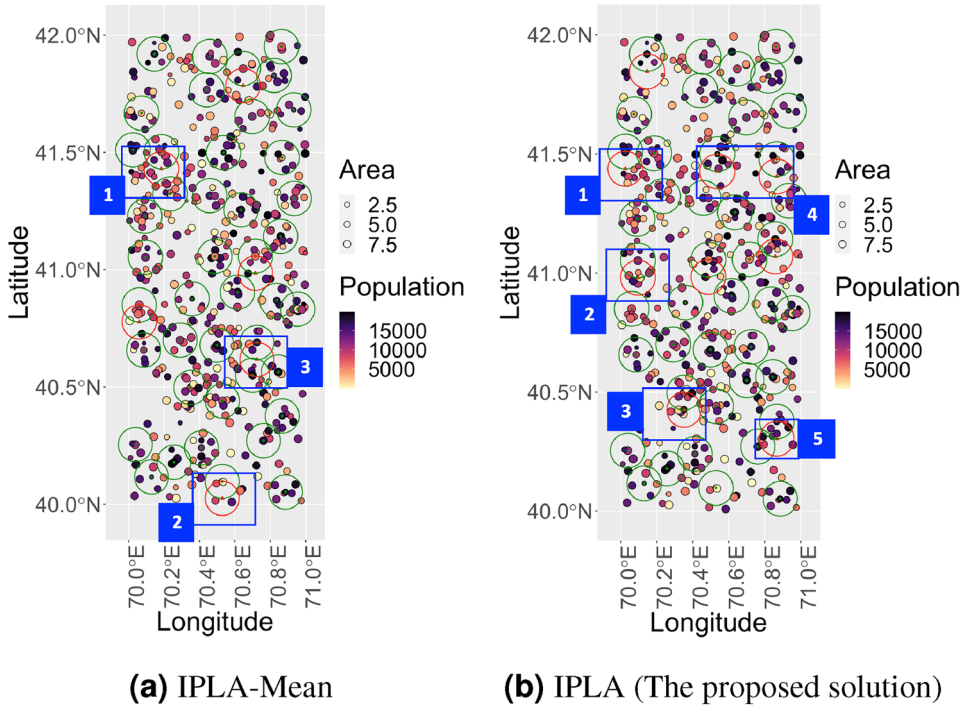


Fig. 5. The new PODs are located five miles away from the existing PODs – ONE state for different approaches. The dots represent the demand points, and the circle frames represent the allocation of demand points to PODs within five miles.

increase restaurants and stores to reach a larger population and increase revenue. However, the weighted distance used in this study may require further refinement.

In future work, extensive experiments on larger datasets will be conducted to validate the scalability and effectiveness of the proposed approach, ensuring its applicability to complex real-world scenarios. Additionally, relevant stakeholders will be engaged to gather user feedback and conduct on-the-ground validation through real-world deployments. This will help ensure that the proposed methodology aligns with user needs and performs effectively in dynamic, real-world environments. Future work will also involve integrating the model into a framework to assist emergency response planners and decision-makers. This framework will evaluate and refine strategies, incorporating dynamic updates such as infection rates and mobility data to enhance real-time decision-making and response effectiveness. Finally, in future work, our analysis will include comparisons with effective optimisers such as NSGA-III, to further validate our approach against state-of-the-art methods.

Data availability

Jeddah data cannot be shared openly but are available on request from authors. Requests for the data used in this study should be addressed to N.G., R.A., or S.A. The synthetic data are available in https://github.com/srisanger/selecting-pharmacies-for-COVID-testing

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Declarations

Competing interests
The authors declare no competing interests.

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